

VisionGuard AI

Business Requirements Document (BRD)

This document defines the complete business requirements for VisionGuard AI — an Enterprise Factory Safety Monitoring and Compliance Platform. It covers all functional requirements, non-functional requirements, security, AI detection, alert management, reporting, and enterprise hierarchy.

Version	v2.0 — Enterprise Edition
Document Type	Business Requirements Document (BRD)
Platform	VisionGuard AI
Status	Final

VisionGuard AI

Business Requirements Document (BRD)

Enterprise Factory Safety Monitoring & Compliance Platform

Version 2.0 — Extended Enterprise Edition

Document Control

Field	Value
Document Title	VisionGuard AI — Business Requirements Document
Version	2.0
Status	Final
Project Name	VisionGuard AI
Project Type	AI-Powered Factory Safety Monitoring Platform
Deployment Model	On-Premise (Cloud-ready)
Platform	Web-Based
Prepared For	Factory Management · Safety Department · IT Department
Target Users	Plant Admin · Safety Officers · Supervisors · Security · Viewers
Initial Capacity	50 Cameras · 1 Factory
Future Capacity	1000+ Cameras · 100+ Factories

Table of Contents

- 1. [Executive Summary](#1-executive-summary)
- 2. [Current Business Challenges](#2-current-business-challenges)
- 3. [Gap Analysis — BRD v1.0 vs Enterprise Edition](#3-gap-analysis)
- 4. [Proposed Solution](#4-proposed-solution)
- 5. [Business Goals & KPIs](#5-business-goals--kpis)
- 6. [System Scope](#6-system-scope)
- 7. [Factory & Organizational Structure](#7-factory--organizational-structure)

8. [User Roles & Permissions](#8-user-roles--permissions)
 9. [Functional Requirements — All Modules](#9-functional-requirements--all-modules)
 10. [Business Rules](#10-business-rules)
 11. [Non-Functional Requirements](#11-non-functional-requirements)
 12. [AI & Detection Requirements](#12-ai--detection-requirements)
 13. [Security Requirements](#13-security-requirements)
 14. [Reporting Requirements](#14-reporting-requirements)
 15. [Notification Requirements](#15-notification-requirements)
 16. [Integration Requirements](#16-integration-requirements)
 17. [Hardware Requirements](#17-hardware-requirements)
 18. [Workflow Scenarios](#18-workflow-scenarios)
 19. [Success Criteria](#19-success-criteria)
 20. [Future Roadmap](#20-future-roadmap)
-

1. Executive Summary

Factory safety monitoring in manufacturing environments is currently a manual, reactive process. Supervisors and safety officers physically patrol production floors to verify PPE compliance, monitor zone occupancy, and identify unauthorized access. This approach is slow, error-prone, produces no audit evidence, and cannot scale.

VisionGuard AI replaces this manual process with an AI-powered, real-time monitoring platform. The system connects to existing factory CCTV cameras via RTSP, analyzes video feeds using computer vision models, detects violations automatically, and alerts the right people instantly — all from a centralized web dashboard.

The platform is designed as a **production-ready enterprise system** — not a demo or prototype. It must support real factory deployments from day one, with a clear scaling path from 1 factory and 50 cameras to 100 factories and 1000+ cameras.

2. Current Business Challenges

2.1 Problem 1 — PPE Non-Compliance

Workers frequently enter production zones without required Personal Protective Equipment.

Affected PPE Types:

- Safety Helmet (Hard Hat)
- Safety Vest (High-visibility)
- Safety Gloves
- Safety Shoes / Steel-toe boots
- Face Mask (in applicable zones)

Current Detection Method: Manual observation by supervisors and safety officers.

Problems with Current Method:

Problem	Impact
Human error in observation	Violations missed regularly
Delayed detection	Worker already in danger zone
No photographic evidence	Disputes during incident investigation
No audit trail	Compliance audits fail
No trend data	Cannot identify repeat offenders or high-risk zones
No automated alerts	Supervisor must be physically present

Business Impact:

- Regulatory fines from labor safety authorities
- Increased workplace injury rate
- Higher insurance premiums
- Failed factory safety audits

2.2 Problem 2 — Zone Overcrowding

Production zones have defined occupancy limits for safety reasons. These limits are regularly exceeded without anyone noticing.

Example Zone Limits:

Zone	Max Capacity	Risk if Exceeded
Welding Area	5	Fire hazard, reduced visibility
Paint Shop	3	Fume inhalation risk
Boiler Area	2	Explosion risk
Chemical Storage	1	Contamination risk
Assembly Line	10	Collision risk
Packaging Area	8	Ergonomic injury risk

Problems:

- No real-time person count per zone
- No automatic alert when limit exceeded
- Safety risk accumulates silently
- No historical data on peak occupancy patterns

2.3 Problem 3 — Delayed Incident Response

Incidents are reported after they occur — often hours later — by which time evidence is lost and response is reactive.

Examples:

- Worker fall — discovered only when worker does not return
- Unauthorized entry — discovered during manual review of footage
- PPE violation — reported after a near-miss incident
- Camera offline — discovered during next scheduled maintenance

Impact:

- Management receives information too late to prevent harm
 - No chain of custody for evidence
 - Compliance investigations lack supporting data
-

2.4 Problem 4 — No Centralized Monitoring

Most factories have existing CCTV systems but:

- No centralized dashboard to view all cameras
 - No AI analytics layer
 - No alert automation
 - No occupancy tracking
 - Multiple disconnected systems (CCTV, access control, HR) with no integration
 - No historical trend analysis
-

2.5 Problem 5 — Compliance & Audit Readiness (Gap — Not in BRD v1.0)

Factories undergo periodic safety audits by regulatory bodies. Currently:

- No structured audit trail of safety events
 - No exportable compliance reports
 - No evidence of corrective actions taken
 - No documentation of alert resolution workflows
-

2.6 Problem 6 — No Safety Performance Measurement (Gap — Not in BRD v1.0)

Factory management has no objective safety KPIs:

- Cannot measure PPE compliance rate over time
- Cannot compare safety performance between zones or shifts
- Cannot identify safety improvement or degradation trends
- No safety score to drive improvement programs

3. Gap Analysis

BRD v1.0 → Enterprise Edition v2.0

Area	BRD v1.0	Enterprise v2.0	Gap Filled
PPE Detection	Helmet + Vest only	Helmet + Vest + Gloves + Shoes + Mask	■
Occupancy	Basic count	Count + peak tracking + heatmap + shift analysis	■
Alerts	Basic alert	Full lifecycle: Open → Ack → InProgress → Closed + history	■
Roles	4 roles	4 roles + granular per-module permissions	■
Notifications	Not specified	Email + In-App + Web Push + Slack + Teams + Webhook	■
Multi-Factory	Single factory	Multi-factory with organization-level isolation	■
Analytics	Basic %	KPIs + trends + heatmaps + safety score + shift analysis	■
Reports	Not specified	PDF + Excel, scheduled + on-demand	■
Security	JWT + RBAC	JWT + refresh rotation + blacklisting + OWASP + audit	■
AI Pipeline	Not specified	YOLO + ByteTrack + Rules Engine + hot-swap	■
Camera Health	Offline alert	FPS + latency + circuit breaker + reconnect policy	■

Area	BRD v1.0	Enterprise v2.0	Gap Filled
Audit Trail	Not specified	Immutable append-only audit log, compliance export	■
Monitoring	Not specified	Prometheus + Grafana + Loki + Alertmanager	■
Scalability	50→200 cameras	50→1000+ cameras, multi-factory, horizontal scaling	■
Rules Engine	Hardcoded	Fully configurable, DB-driven, duration-aware	■
Deployment	Not specified	Docker + Nginx + CI/CD + prod compose	■
Integrations	Listed as future	API-ready stubs for SAP, HRMS, SMS, Fire Alarm	■
Worker Tracking	Not specified	AI worker registry, heartbeat, camera assignment	■

4. Proposed Solution

VisionGuard AI will:

1. Connect to all factory RTSP cameras
2. Process video streams in real time using AI (YOLO + ByteTrack)
3. Detect PPE violations, overcrowding, and unauthorized entry automatically
4. Evaluate violations against configurable business rules
5. Generate alerts and route them to the right supervisors
6. Store violation snapshots as evidence
7. Provide a real-time web dashboard for all stakeholders
8. Maintain complete historical records for compliance
9. Provide analytics, trends, and safety KPIs
10. Send notifications via multiple channels instantly
11. Generate compliance reports on demand
12. Support scaling from 1 to 100+ factories without re-architecture

5. Business Goals & KPIs

Business Goals

Goal	Metric	Target
Reduce PPE violations	PPE violation count	-80% within 6 months
Reduce safety incidents	Incident count per month	-50% within 12 months
Improve alert response	Alert acknowledgement time	< 5 minutes
Improve audit readiness	Audit findings	Zero evidence gaps
Real-time monitoring	Dashboard latency	< 5 seconds
System reliability	Camera uptime monitoring	99.5% availability

Operational KPIs

KPI	Formula	Frequency
Helmet Compliance %	$(\text{Compliant frames} / \text{Total person frames}) \times 100$	Daily
Vest Compliance %	$(\text{Compliant frames} / \text{Total person frames}) \times 100$	Daily
Overall PPE Compliance %	Average of all PPE KPIs	Daily
Safety Score	$0.6 \times \text{PPE\%} + 0.3 \times (1 - \text{incident rate}) + 0.1 \times \text{closure speed}$	Weekly
Alert Response Time	$\text{avg}(\text{acknowledged_on} - \text{created_on})$	Daily
Alert Closure Time	$\text{avg}(\text{resolved_on} - \text{created_on})$	Daily
Peak Occupancy	$\text{max}(\text{current_count})$ per zone per day	Daily
Overcrowding Frequency	Count of overcrowding events per zone	Weekly
Camera Uptime %	$(\text{Online minutes} / \text{Total minutes}) \times 100$	Daily
Incident Rate	Incidents per 1000 person-hours	Monthly

6. System Scope

In Scope

- Real-time RTSP camera stream processing

- PPE detection: Helmet, Vest, Gloves, Shoes, Mask
- Occupancy monitoring and zone capacity enforcement
- Unauthorized zone entry detection
- Camera health monitoring (online/offline/FPS/latency)
- Configurable business rules and alert routing
- Multi-channel notification delivery
- Real-time web dashboard with WebSocket push
- Historical analytics and trend reporting
- Compliance report generation (PDF + Excel)
- Immutable audit trail
- Role-based access control
- Multi-factory, multi-organization support
- AI model versioning and hot-swap
- Docker-based deployment

Out of Scope (Phase 1)

- Face recognition / worker identity
 - Mobile application (native iOS/Android)
 - Edge AI processing (on-camera inference)
 - Fire and smoke detection
 - Fall detection
 - Forklift/vehicle detection
 - Predictive analytics / AI risk scoring
 - SAP / HRMS integration (API stubs only)
 - Biometric access control
-

7. Factory & Organizational Structure

Hierarchy

```
Organization (Tenant)
  ■■■ Factory (Plant)
    ■■■ Zone (Production Area)
      ■■■ Camera (RTSP Stream)
```

Example

```
Organization: {Enterprise Name}

■■■ Factory: {Factory Name 1}
  ■ ■■■ Zone: Assembly Area (Max: 10 people) [3 cameras]
```

- ■■■ Zone: Welding Area (Max: 5 people) [2 cameras]
- ■■■ Zone: Paint Shop (Max: 3 people) [2 cameras]
- ■■■ Zone: Packaging Area (Max: 8 people) [2 cameras]
- ■■■ Zone: Warehouse (Max: 15 people) [3 cameras]
- ■■■ Zone: Dispatch Area (Max: 6 people) [2 cameras]
- ■■■ Zone: Chemical Storage (Max: 1 person) [1 camera]
- ■■■ Zone: Boiler Area (Max: 2 people) [1 camera]
- ■■■ Zone: Loading Dock (Max: 4 people) [2 cameras]
-
- Factory: {Factory Name 2}
- Zone: CNC Machining (Max: 8 people) [2 cameras]
- Zone: Quality Control (Max: 6 people) [2 cameras]

8. User Roles & Permissions

Role Definitions

Role 1 — Plant Admin

Scope: Full access to assigned factories.

Responsibilities:

- Manage factory structure (zones, cameras)
- Manage users and assign roles
- Configure detection rules per zone
- View all dashboards, alerts, analytics
- Generate and export compliance reports
- Access audit logs
- Manage AI worker assignments
- Configure notification templates

Permissions:

Module	Create	Read	Update	Delete
Factory	■	■	■	■
Zone	■	■	■	■
Camera	■	■	■	■
Users	■	■	■	■
Alerts	■	■	■	■
Config	■	■	■	■
Reports	■	■	—	—

Module	Create	Read	Update	Delete
Audit	—	■	—	—
Analytics	—	■	—	—

Role 2 — Safety Officer

Scope: All zones in assigned factory.

Responsibilities:

- Monitor all alerts and violations
- Investigate incidents with snapshot evidence
- Acknowledge and resolve alerts
- Generate safety compliance reports
- View analytics and trends
- View audit logs for their actions

Permissions:

Module	Create	Read	Update	Delete
Factory	—	■	—	—
Zone	—	■	—	—
Camera	—	■	—	—
Alerts	—	■	■ (ack/resolve)	—
Reports	■	■	—	—
Analytics	—	■	—	—
Audit	—	■ (own actions)	—	—

Role 3 — Supervisor

Scope: Assigned zones only.

Responsibilities:

- Monitor assigned zones in real time
- Acknowledge and resolve alerts for assigned zones
- Take corrective actions and log comments
- View zone-level dashboard

Permissions:

Module	Create	Read	Update	Delete
Zone (assigned)	—	■	—	—
Camera (assigned)	—	■	—	—
Alerts (assigned zones)	—	■	■ (ack/resolve)	—
Analytics (zone)	—	■	—	—

Role 4 — Viewer

Scope: Read-only access to assigned dashboards.

Responsibilities:

- View dashboards assigned by admin
- No alert management capability
- Typically used for plant managers, executives, auditors

Permissions:

Module	Create	Read	Update	Delete
Dashboard	—	■	—	—
Alerts	—	■	—	—
Analytics	—	■	—	—

9. Functional Requirements — All Modules

Module 1 — Identity & Access Management

1.1 User Registration & Management

FR-IAM-001: Admin shall be able to create users with the following fields:

- Full Name
- Email Address (unique, used as login)
- Role (Admin / Safety Officer / Supervisor / Viewer)
- Assigned Factory
- Assigned Zones (for Supervisor role)
- Status (Active / Inactive)

FR-IAM-002: Admin shall be able to activate, deactivate, or suspend user accounts.

FR-IAM-003: System shall send a welcome email to newly created users with a one-time password setup link.

FR-IAM-004: Users shall be able to update their own profile (name, password, notification preferences).

1.2 Authentication

FR-IAM-005: System shall authenticate users via email + password.

FR-IAM-006: System shall issue a JWT Access Token (8-hour expiry) and a Refresh Token (7-day expiry) on successful login.

FR-IAM-007: System shall implement Refresh Token Rotation — each refresh issues a new token and invalidates the old one.

FR-IAM-008: System shall blacklist Access Tokens on logout using Redis TTL.

FR-IAM-009: System shall detect and prevent Refresh Token replay attacks — reuse of a revoked token triggers full session invalidation.

FR-IAM-010: System shall support password reset via email OTP.

FR-IAM-011: System shall enforce password policy: minimum 8 characters, at least 1 uppercase, 1 number, 1 special character.

FR-IAM-012: System shall lock account after 5 consecutive failed login attempts for 15 minutes.

1.3 Authorization

FR-IAM-013: System shall enforce Role-Based Access Control (RBAC) on every API endpoint.

FR-IAM-014: Unauthorized access attempts shall return HTTP 403 and be logged to audit.

FR-IAM-015: Permissions shall be stored per role as JSONB for flexibility.

FR-IAM-016: Supervisors shall only access alerts and dashboards for their assigned zones.

Module 2 — Organization Management

2.1 Organization

FR-ORG-001: System shall support multi-tenant organizations.

FR-ORG-002: Each organization shall have complete data isolation from other organizations.

FR-ORG-003: Organization fields:

- Organization Name
- Organization Code (unique)
- Industry Type
- Contact Person
- Contact Email
- Status (Active / Inactive)
- Subscription Plan (future)

FR-ORG-004: System Admin (super-user) shall manage organizations. Organization-level Admin manages factories within their org.

Module 3 — Factory Management

3.1 Factory CRUD

FR-FAC-001: Admin shall create factories with:

- Factory Name
- Factory Code (unique within org)
- Address
- City / State / Country
- Plant Head (linked user)
- GPS Coordinates (optional, for reporting)
- Status (Active / Inactive)

FR-FAC-002: Admin shall edit factory details and status.

FR-FAC-003: Deleting a factory shall be a soft delete — all historical data retained.

FR-FAC-004: Factory list shall be searchable, filterable by status, and paginated.

FR-FAC-005: Factory dashboard shall show:

- Total zones
 - Total cameras (online / offline)
 - Active alerts count
 - Today's violations count
 - PPE compliance % (today)
 - Safety score
-

Module 4 — Zone Management

4.1 Zone CRUD

FR-ZON-001: Admin shall create zones within a factory with:

- Zone Name
- Zone Code (unique within factory)
- Description
- Max Occupancy (integer)
- Supervisor (linked user)
- Zone Type (Production / Storage / Restricted / Entry-Exit / Office)
- Restricted Zone flag (boolean — triggers unauthorized entry alerts)
- Status (Active / Inactive)

FR-ZON-002: Admin shall assign multiple cameras to a zone.

FR-ZON-003: Admin shall assign a supervisor to a zone. Supervisors receive alerts only for their assigned zones.

FR-ZON-004: Zone list shall show current occupancy vs max occupancy with color indicators:

- Green: < 70% capacity
- Yellow: 70–90% capacity
- Red: > 90% capacity or exceeded

FR-ZON-005: Soft delete only — zones with historical data cannot be hard deleted.

4.2 Zone Configuration

FR-ZON-006: Admin shall configure per-zone detection settings:

- Person detection confidence threshold
- Helmet detection confidence threshold
- Vest detection confidence threshold
- Gloves detection confidence threshold
- Shoes detection confidence threshold
- Mask detection confidence threshold
- Required PPE list (checkboxes)
- Max occupancy override (same as FR-ZON-001 but can be updated live)
- Frame sampling rate (FPS to process)
- Violation cooldown period (minimum seconds between same violation alerts)

FR-ZON-007: Config changes shall take effect in AI workers within 10 seconds without restarting streams.

FR-ZON-008: Config version number shall increment on every save (used for race-condition-safe sync with AI workers).

Module 5 — Camera Management

5.1 Camera CRUD

FR-CAM-001: Admin shall register cameras with:

- Camera Name
- Camera Code (unique within factory)
- RTSP URL (encrypted at rest)
- Zone assignment
- Camera Type (Fixed / PTZ / Fisheye)
- Position Description (e.g., "North wall, facing main entry")
- Status (Active / Inactive)

FR-CAM-002: Admin shall update camera details including RTSP URL.

FR-CAM-003: System shall validate RTSP URL format before saving.

FR-CAM-004: Camera list shall show:

- Camera name and code
- Zone assignment
- Online / Offline / Degraded status
- Current FPS
- Last seen timestamp
- Assigned AI worker

5.2 Camera Health Monitoring

FR-CAM-005: System shall monitor each camera's health in real time:

- Online / Offline status
- Stream FPS (live)
- Stream latency
- Last frame timestamp
- Reconnection attempt count

FR-CAM-006: Camera shall be marked Offline if no frame received for 60 seconds.

FR-CAM-007: Camera shall be marked Degraded if FPS drops below 50% of expected rate.

FR-CAM-008: AI workers shall attempt automatic reconnection:

- Retry 1: after 5 seconds
- Retry 2: after 15 seconds
- Retry 3: after 60 seconds
- After 3 failures: publish CameraOfflineDetected event

FR-CAM-009: Circuit breaker: 3 consecutive processing failures on same camera → camera isolated → CameraOfflineDetected published.

Module 6 — AI Worker Management

6.1 Worker Registry

FR-WRK-001: System shall maintain a registry of all AI worker instances with:

- Worker ID (unique)
- Worker Name
- Host IP / Hostname
- Status (Online / Offline / Degraded)
- Assigned cameras (list)
- Current model version
- GPU / CPU mode
- Last heartbeat timestamp
- Frames processed (total)

FR-WRK-002: Workers shall publish heartbeat events every 30 seconds.

FR-WRK-003: Worker shall be marked offline if no heartbeat received for 90 seconds.

FR-WRK-004: Admin shall view worker health dashboard showing all workers and their assigned cameras.

FR-WRK-005: Admin shall be able to trigger camera reassignment between workers (future Phase 2 — auto-rebalancing).

6.2 Model Management

FR-WRK-006: Admin shall upload new AI model versions via the backend API.

FR-WRK-007: New model versions shall be stored in MinIO model registry.

FR-WRK-008: Backend shall publish ModelUpdated event to trigger hot-swap on all workers.

FR-WRK-009: Workers shall swap the model without restarting camera streams.

FR-WRK-010: Each worker shall report its active model version in heartbeat events.

Module 7 — Occupancy Monitoring

7.1 Real-Time Occupancy

FR-OCC-001: System shall count persons in each zone in real time using AI detection + ByteTrack tracking.

FR-OCC-002: Occupancy count shall be updated every time a detection frame is processed (default every 500ms at 2 FPS).

FR-OCC-003: Dashboard shall show live occupancy per zone updated via WebSocket.

FR-OCC-004: Occupancy gauge shall use color coding:

- Green: 0 to 70% of max

- Yellow: 70% to 100% of max
- Red: Exceeded max

FR-OCC-005: IF `current_count` > `max_occupancy` THEN generate `OvercrowdingDetected` event → create High severity alert.

FR-OCC-006: Overcrowding alert shall not re-trigger within a configurable cooldown period (default: 2 minutes).

7.2 Occupancy History

FR-OCC-007: System shall log occupancy readings every minute to `occupancy_logs`.

FR-OCC-008: System shall track peak occupancy per zone per day.

FR-OCC-009: Analytics shall show:

- Average occupancy by hour (heatmap)
- Peak occupancy by day
- Overcrowding frequency by zone
- Occupancy trend (7-day, 30-day)

Module 8 — PPE Detection

8.1 Detection Types

FR-PPE-001: Helmet Detection

System shall detect presence or absence of safety helmet on each detected person.

Detection states:

- Helmet Present ■
- Helmet Missing ■

Rule:

```
IF person_detected AND helmet_missing
FOR >= 3 seconds (sustained violation)
THEN create violation + store snapshot + generate alert
```

FR-PPE-002: Safety Vest Detection

System shall detect presence or absence of high-visibility safety vest.

Detection states:

- Vest Present ■
- Vest Missing ■

Rule:

```
IF person_detected AND vest_missing
FOR >= 3 seconds
THEN create violation + store snapshot + generate alert
```

FR-PPE-003: Gloves Detection

System shall detect presence or absence of safety gloves.

Applicable zones: Welding Area, Chemical Storage, CNC Machining.

Rule:

```
IF zone.requires_gloves AND person_detected AND gloves_missing
FOR >= 5 seconds
THEN create violation + store snapshot + generate alert
```

FR-PPE-004: Safety Shoes Detection

System shall detect presence or absence of safety shoes / steel-toe boots.

Rule:

```
IF zone.requires_shoes AND person_detected AND shoes_missing
FOR >= 5 seconds
THEN create violation + store snapshot + generate alert
```

FR-PPE-005: Face Mask Detection

System shall detect presence or absence of face mask.

Applicable zones: Paint Shop, Chemical Storage.

Rule:

```
IF zone.requires_mask AND person_detected AND mask_missing
FOR >= 3 seconds
THEN create violation + store snapshot + generate alert
```

8.2 Detection Pipeline

```
RTSP Frame
↓
Frame Sampling (configurable FPS)
↓
Person Detection (YOLO)
↓
Confidence Threshold Check
↓
ByteTrack – assign Track IDs
↓
For each tracked person:
```

```
Helmet Check → if zone.requires_helmet
Vest Check → if zone.requires_vest
Gloves Check → if zone.requires_gloves
Shoes Check → if zone.requires_shoes
Mask Check → if zone.requires_mask
↓
Rules Engine – evaluate duration + cooldown
↓
Violation Created (if rule fires)
↓
Snapshot Captured → MinIO
↓
Event Published → RabbitMQ
```

8.3 Violation Record

Each violation stores:

- Violation ID (UUID)
- Zone ID
- Camera ID
- Violation Type (Enum)
- Confidence Score
- ByteTrack Track ID
- Snapshot Key (MinIO object path)
- Timestamp
- Rules Engine Rule ID that fired

Module 9 — Alert Management

9.1 Alert Categories & Severity

Category	Alert Type	Severity	Auto-Assign To
PPE	Helmet Missing	High	Zone Supervisor
PPE	Vest Missing	Medium	Zone Supervisor
PPE	Gloves Missing	Medium	Zone Supervisor
PPE	Shoes Missing	Medium	Zone Supervisor
PPE	Mask Missing	Medium	Zone Supervisor
Occupancy	Overcrowding	High	Zone Supervisor
Access	Unauthorized Entry	High	Security + Supervisor
Infrastructure	Camera Offline	High	IT Admin

Category	Alert Type	Severity	Auto-Assign To
Infrastructure	Camera Degraded (low FPS)	Low	IT Admin
Infrastructure	Worker Offline	High	IT Admin
Safety	Multiple Violations Same Person	Critical	Safety Officer

9.2 Alert Lifecycle

```
[OPEN]
↓ Supervisor acknowledges
[ACKNOWLEDGED]
↓ Supervisor starts action
[IN PROGRESS]
↓ Supervisor confirms corrective action taken
[CLOSED]
```

Every status transition is recorded in `alerts.alert_history` with:

- From Status
- To Status
- Changed By (user)
- Changed At (timestamp)
- Comment (mandatory on Close)

9.3 Alert Fields

- Alert ID (UUID)
- Alert Number (human-readable: ALT-2026-0001)
- Violation ID (FK, nullable for infra alerts)
- Factory ID
- Zone ID
- Camera ID
- Alert Type
- Severity (Critical / High / Medium / Low)
- Status (Open / Acknowledged / InProgress / Closed)
- Assigned To (user)
- Snapshot URL (pre-signed MinIO URL)
- Created On
- Acknowledged On
- Resolved On
- SLA Due At (based on severity SLA config)

9.4 Alert SLA

Configurable per severity:

Severity	Acknowledgement SLA	Resolution SLA
Critical	2 minutes	15 minutes
High	5 minutes	30 minutes
Medium	15 minutes	2 hours
Low	1 hour	8 hours

SLA breach shall trigger escalation notification to Safety Officer and Admin.

9.5 Alert Deduplication

System shall not create a new alert for the same violation type on the same camera if:

- An Open or Acknowledged alert of the same type already exists for that zone
- AND the cooldown period has not elapsed

Configurable cooldown period per violation type per zone.

9.6 Alert Filters

Alert list page shall support filtering by:

- Date range
- Factory
- Zone
- Camera
- Severity
- Status
- Violation type
- Assigned to
- SLA status (within SLA / breached)

Module 10 — Incident Management

10.1 Incident vs Alert

Every alert of severity High or Critical automatically becomes an Incident.

Field	Value
Incident Number	INC-2026-0001 (auto-generated)
Linked Alert	Alert ID
Incident Type	Derived from alert type
Zone	From alert

Field	Value
Camera	From alert
Timestamp	From alert
Severity	From alert
Status	Open / Under Investigation / Corrective Action / Closed
Assigned To	Safety Officer
Evidence	Snapshot + video clip reference (future)
Root Cause	Text (filled by Safety Officer)
Corrective Action	Text (mandatory on close)
Closed By	User
Closed At	Timestamp

10.2 Incident Workflow

```
High/Critical Alert Created
↓
Incident Auto-Created (linked to alert)
↓
Safety Officer assigned
↓
Safety Officer reviews snapshot + occupancy data
↓
Safety Officer fills root cause + corrective action
↓
Incident Closed
↓
Alert auto-resolved (if still open)
↓
Audit entry created
```

Module 11 — Dashboard

11.1 Executive Dashboard (Admin / Plant Manager / Viewer)

Real-time metrics (WebSocket):

- Total Cameras / Online / Offline
- Active Alerts by severity (Critical / High / Medium / Low)
- Open Incidents
- PPE Compliance % (today)
- Overall Safety Score
- Violations today (count + trend vs yesterday)

Charts:

- Violations by type (donut chart — today)
 - Alert volume trend (last 7 days — bar chart)
 - PPE compliance trend (last 30 days — line chart)
 - Zone compliance heatmap (grid: zone × PPE type)
-

11.2 Safety Dashboard (Safety Officer)

- Violations by zone (today)
 - Open alerts with SLA countdown
 - Alert resolution time trend
 - Top 5 repeat violation zones
 - Shift-wise violation breakdown (if shift config enabled)
 - Weekly violation trend
-

11.3 Zone Dashboard (Supervisor)

- Assigned zone occupancy gauge (real-time)
 - Assigned zone cameras — live status
 - Active alerts for assigned zones
 - Recent violations (last 10)
 - Zone safety score
-

11.4 Camera Dashboard

- Camera grid view (all cameras, status card per camera)
 - Filter by: zone / status / worker
 - Each card shows: camera name, zone, status, FPS, last seen
 - Click → camera detail: full health info + recent violations
-

11.5 Worker Dashboard (Admin)

- AI worker list: status, assigned cameras, model version, heartbeat
 - Worker health: frames processed per minute, inference latency (p50/p99)
 - DLQ depth per worker
 - Model version distribution
-

Module 12 — Analytics

12.1 PPE Analytics

- Helmet compliance % — daily / weekly / monthly trend
- Vest compliance % — daily / weekly / monthly trend
- Violations by type — bar chart
- Violations by zone — horizontal bar
- Violations by hour of day — heatmap (zone × hour)
- Top 10 violation zones (ranking)
- Violation trend (last 90 days)

12.2 Occupancy Analytics

- Average occupancy per zone — bar chart
- Peak occupancy per zone per day — calendar heatmap
- Overcrowding events per zone — trend
- Occupancy by hour (heatmap: zone × hour)
- Zone utilization % (avg occupancy / max occupancy)

12.3 Alert Analytics

- Alert volume by day (last 30 days)
- Alert volume by severity
- Average acknowledgement time by role
- Average resolution time by zone
- SLA breach rate
- Alert escalation rate
- Top 5 alert types

12.4 Safety Score

Calculated weekly per factory and per zone.

```
Safety Score = (0.6 × PPE Compliance%) +  
(0.3 × (1 - normalized_incident_rate)) +  
(0.1 × alert_closure_speed_score)
```

Where:

```
normalized_incident_rate = incidents / (person_hours / 1000)  
alert_closure_speed_score = 1 - (avg_closure_time / SLA_target_time)
```

Score displayed as:

- 90–100: Excellent (Green)
- 70–89: Good (Yellow)
- 50–69: Needs Improvement (Orange)
- < 50: Critical (Red)

Module 13 — Reports

13.1 Report Types

Report	Audience	Format	Frequency
Daily Safety Summary	Admin / Safety Officer	PDF	Daily (auto)
Weekly Compliance Report	Admin / Plant Manager	PDF + Excel	Weekly (auto)
Monthly Safety Scorecard	Management	PDF	Monthly (auto)
Alert Detail Report	Safety Officer	PDF + Excel	On-demand
Violation Evidence Report	Safety Officer / Auditor	PDF	On-demand
Camera Health Report	IT Admin	Excel	On-demand
Audit Trail Export	Admin	PDF + Excel	On-demand
Occupancy Trend Report	Plant Manager	Excel	On-demand

13.2 Report Fields (Daily Safety Summary)

- Report period
- Factory name
- Total violations (by type)
- PPE compliance %
- Active alerts at end of day
- Alerts resolved today
- Camera uptime %
- Safety score
- Top 3 violation zones
- Escalated incidents

13.3 Scheduled Reports

- Admin configures report schedule (daily / weekly / monthly)
- System generates and emails PDF to configured recipients
- Reports stored in MinIO and accessible from Reports page

Module 14 — Configuration (Rules Engine)

14.1 Zone Detection Config

Per-zone configurable settings (stored in `config.zone_configs`):

Setting	Type	Default
person_threshold	DECIMAL	0.70
helmet_threshold	DECIMAL	0.75

Setting	Type	Default
vest_threshold	DECIMAL	0.75
gloves_threshold	DECIMAL	0.70
shoes_threshold	DECIMAL	0.70
mask_threshold	DECIMAL	0.75
max_occupancy	INT	From zone
frame_sample_fps	INT	2
ppe_required	JSONB	["helmet","vest"]
restricted_zone	BOOLEAN	false

14.2 Rules Engine Configuration

Each rule is stored in `config.zone_rules` and is fully configurable:

Field	Type	Description
rule_id	UUID	
zone_id	UUID	
rule_name	VARCHAR	Human-readable name
condition_type	ENUM	HELMET_MISSING, VEST_MISSING, etc.
duration_seconds	INT	Sustained violation before alert fires
cooldown_seconds	INT	Min time between same alerts
severity	ENUM	
enabled	BOOLEAN	
actions	JSONB	["CREATE_ALERT","STORE_SNAPSHOT","NOTIFY"]
notify_roles	JSONB	["Supervisor","SafetyOfficer"]
notify_channels	JSONB	["Email","InApp","WebPush"]

14.3 Alert Routing Configuration

Admin configures who receives alerts per violation type per zone:

- Primary assignee (default: zone supervisor)
- Escalation assignee (default: safety officer)
- Escalation after: configurable SLA per severity

14.4 Notification Templates

Templates stored in DB, configurable per organization:

- Template name
- Channel (Email / Slack / Teams / Webhook)
- Subject template (with placeholders)
- Body template (with placeholders)
- Available placeholders: `{zone_name}`, `{camera_name}`, `{violation_type}`, `{timestamp}`, `{alert_url}`, `{snapshot_url}`

Module 15 — Notifications

15.1 Notification Channels

Channel	Use Case	Configuration
Email	Alert notifications + reports	SMTP server config
In-App	Real-time alert feed	Built-in (WebSocket)
Web Push	Browser push notifications	Web Push API keys
Slack	Team channel alerts	Slack Webhook URL
Microsoft Teams	Enterprise team alerts	Teams Webhook URL
Webhook	Custom integrations	HTTP POST URL + auth

15.2 User Notification Preferences

Each user configures preferred channels per alert severity:

Severity	User Choice
Critical	Email + In-App + Web Push
High	Email + In-App
Medium	In-App
Low	In-App

15.3 Notification Delivery

- All notifications logged to `notifications.notification_log`
- Failed delivery: retry 3 times with exponential backoff
- All retries failed: escalate to admin + log to audit
- Delivery status tracked: Sent / Delivered / Failed

Module 16 — Audit Trail

16.1 Audited Actions

Every sensitive action is recorded:

Action	Trigger
UserCreated	New user account
UserUpdated	Profile or role changed
UserDisabled	Account deactivated
LoginSuccess	Successful login
LoginFailed	Failed login attempt
Logout	Session terminated
AlertAcknowledged	Supervisor acknowledges
AlertResolved	Alert closed
AlertReassigned	Alert reassigned to different user
IncidentClosed	Incident closed with root cause
ConfigChanged	Zone config or rule changed
CameraAdded	New camera registered
CameraRemoved	Camera deleted
ModelUpdated	AI model version changed
UnauthorizedAccess	403 response on any endpoint
ReportGenerated	Report created
UserPasswordReset	Password reset performed

16.2 Audit Log Fields

- ID (UUID)
- User ID
- Action (from above list)
- Entity Type (User / Alert / Camera / Config / ...)
- Entity ID
- Old Value (JSONB — before state)
- New Value (JSONB — after state)
- IP Address
- User Agent
- Correlation ID (request ID)
- Timestamp

16.3 Audit Log Rules

- Append-only: no UPDATE or DELETE permitted at database level
- Accessible only by Admin role
- Exportable to PDF or Excel for compliance
- Retained for 36 months online, 5 years cold archive

Module 17 — Health & System Status

17.1 Health Endpoints

Endpoint	Purpose
GET /health/live	Liveness check — is process running?
GET /health/ready	Readiness check — can it serve traffic?
GET /health/status	Full system status (DB, Redis, RabbitMQ, MinIO)

17.2 Readiness Checks

System is ready only when:

- PostgreSQL: connected and responding
- Redis: connected and responding
- RabbitMQ: connected and queue accessible
- MinIO: connected and bucket accessible

17.3 System Status Response

```
{
  "status": "healthy",
  "timestamp": "2026-01-01T10:00:00Z",
  "components": {
    "database": { "status": "up", "latency_ms": 2 },
    "redis": { "status": "up", "latency_ms": 1 },
    "rabbitmq": { "status": "up", "queue_depth": 0 },
    "minio": { "status": "up", "latency_ms": 5 },
    "ai_workers": {
      "total": 4,
      "online": 4,
      "offline": 0
    }
  }
}
```

10. Business Rules

Rule ID	Rule	Notes
BR-001	Camera must be assigned to a zone before activation	Enforced at API level
BR-002	Zone max occupancy cannot be 0	Min value: 1
BR-003	Alert deduplication: same violation type + same zone → no new alert during cooldown	Cooldown configurable
BR-004	Alert SLA breach → auto-escalation to Safety Officer	Configurable per severity
BR-005	Audit log is immutable — no update or delete	DB-level enforcement
BR-006	Supervisor can only access alerts for their assigned zones	RBAC enforcement
BR-007	Violation snapshot must be stored before alert is created	Atomicity required
BR-008	Config version must increment on every zone config save	For race-condition-safe worker sync
BR-009	AI workers must not access the database directly	Architecture constraint
BR-010	Refresh token reuse after revocation → full session invalidation	Security rule
BR-011	Account locked after 5 failed logins for 15 minutes	Security rule
BR-012	Password must meet complexity policy	Security rule
BR-013	Incident closure requires root cause + corrective action text	Mandatory fields
BR-014	Report download links expire after 24 hours	Security rule
BR-015	MinIO snapshot pre-signed URLs expire in 15 minutes	Security rule

11. Non-Functional Requirements

Performance

Requirement	Target
Alert latency (detection → notification)	≤ 5 seconds
API response time (p95)	≤ 200ms
Dashboard load time (initial)	≤ 2 seconds
WebSocket push latency	≤ 1 second
Report generation time	≤ 10 seconds
Camera offline detection	≤ 60 seconds
Config update propagation to workers	≤ 10 seconds
Inference latency per frame (GPU)	≤ 50ms
Inference latency per frame (CPU)	≤ 500ms

Capacity

Requirement	Initial	Future
Cameras	50	1000+
Factories	1	100+
Concurrent users	50	500+
Concurrent WebSocket connections	100	1000+
Events per second (RabbitMQ)	50	5000+
Violations stored per day	10,000	1,000,000

Availability

Requirement	Target
System availability	99.5%
Planned downtime window	Saturday 2–4 AM
Recovery Time Objective (RTO)	30 minutes
Recovery Point Objective (RPO)	5 minutes

Reliability

- Zero data loss on AI worker crash (Dead Letter Queue)
- Zero event loss on backend restart (RabbitMQ durable queues)

- Automatic camera stream reconnection
- Circuit breaker per camera

Scalability

- Horizontal scaling: add AI workers, add backend instances — no code change
- Database: partition by month, read replicas at scale
- Cache: Redis cache reduces DB load by 80%+
- WebSocket: Redis Pub/Sub for cross-instance broadcast

12. AI & Detection Requirements

Requirement	Target
Helmet detection accuracy	$\geq 90\%$
Vest detection accuracy	$\geq 88\%$
Person detection accuracy	$\geq 95\%$
Occupancy count accuracy	$\geq 95\%$
False positive rate	$\leq 5\%$
False negative rate	$\leq 10\%$

Model Requirements

- Model format: ONNX (for runtime flexibility) + PyTorch (for training)
- CUDA support: NVIDIA GPU acceleration
- CPU fallback: automatic when no GPU detected
- Model versioning: stored in MinIO, versioned by date + hash
- Hot-swap: model update without stream restart

Frame Processing Requirements

- Default sampling: 2 FPS from 25 FPS stream (configurable per zone)
- Maximum detection lag: 500ms from frame capture to event publish
- Snapshot resolution: minimum 640×480, saved as JPEG (quality 85)

13. Security Requirements

Category	Requirement
Authentication	JWT Access Token (8h) + Refresh Token Rotation (7d)
Token Security	Blacklisting on logout, replay attack prevention
Authorization	RBAC enforced on every endpoint
Transport	HTTPS enforced in production (TLS 1.2+)
Data Encryption	RTSP URLs encrypted at rest (AES-256)
Input Validation	Pydantic v2 strict validation on all inputs
SQL Injection	SQLAlchemy ORM — no raw SQL
XSS	React escaping + CSP headers
CSRF	SameSite=Strict cookies + token header for API
Rate Limiting	Per user per role (Redis sliding window)
File Upload	JPEG/PNG only, max 5MB, MIME type validation
Audit	All sensitive actions logged immutably
Secrets	Environment variables only, never committed
OWASP	Top 10 mitigations implemented
Account Lockout	5 failed attempts → 15 min lockout
Password Policy	Min 8 chars, uppercase, number, special char

14. Reporting Requirements

Requirement	Detail
PDF generation	Server-side using WeasyPrint / ReportLab
Excel generation	openpyxl
Max report size	50MB
Report storage	MinIO (auto-deleted after 30 days)
Download link expiry	24 hours
Scheduled reports	Configurable: daily / weekly / monthly
Email delivery	PDF attached to scheduled report email
Charts in PDF	Server-rendered charts (matplotlib / plotly)

15. Notification Requirements

Channel	Library	SLA
Email	SMTP / SendGrid	≤ 30 seconds
In-App	WebSocket	≤ 1 second
Web Push	Web Push API	≤ 5 seconds
Slack	Slack Webhook	≤ 10 seconds
Teams	Teams Webhook	≤ 10 seconds
Webhook	HTTP POST	≤ 10 seconds

Retry policy: 3 attempts with 30s / 2min / 10min backoff.

All notifications logged with delivery status.

16. Integration Requirements

System	Type	Priority	Notes
Email / SMTP	Native	Phase 1	Required for notifications
Slack	Webhook	Phase 1	Alert channel
Microsoft Teams	Webhook	Phase 1	Enterprise teams
SMS Gateway	API	Phase 2	Twilio / local provider
SAP	REST API	Phase 3	Work order creation
HRMS	REST API	Phase 2	Worker identity linkage
Attendance System	REST API	Phase 2	Auto badge → person link
Access Control	REST API	Phase 3	Badge vs camera cross-ref
Fire Alarm System	REST API	Phase 3	Correlated alerts

17. Hardware Requirements

Development Environment

Component	Specification
CPU	Intel Core Ultra 7 165H (or equivalent 8+ core)
RAM	32–64 GB
GPU	NVIDIA RTX 2000 Ada (or equivalent)
Storage	512 GB SSD
OS	Windows 11 + WSL2 / Ubuntu 22.04

Production — 50 Cameras (Initial)

Server	Role	Spec
Server 1	App + DB + Cache + Queue	24-core CPU, 64 GB RAM, 2 TB SSD
Server 2	GPU Worker 1 (25 cameras)	8-core CPU, 32 GB RAM, RTX 4090, 1 TB SSD
Server 3	GPU Worker 2 (25 cameras)	8-core CPU, 32 GB RAM, RTX 4090, 1 TB SSD

Production — 200 Cameras

Server	Role	Count
App Servers	FastAPI backend (load balanced)	2
DB Server	PostgreSQL primary + replica	2
Cache/Queue	Redis + RabbitMQ	1
Storage	MinIO	1
GPU Workers	25 cameras each	8
Monitoring	Prometheus + Grafana + Loki	1

18. Workflow Scenarios

Scenario 1 — Helmet Missing Detection

Worker enters Welding Area without helmet.
Camera detects person via YOLO.

ByteTrack assigns Track ID = T-042.
PPE validator: helmet_detected = False.
Rules engine: helmet_missing FOR 3 seconds → rule fires.
Snapshot captured → uploaded to MinIO.
Event published: HelmetMissingDetected (zone=ZONE-WELDING, camera=CAM-003).
Backend consumer receives event.
Violation record created.
Alert created: ALT-2026-0087, Severity=High, Status=Open.
Incident created: INC-2026-0031 (linked to alert).
Redis cache invalidated.
WebSocket broadcast → supervisor dashboard updated.
Notification dispatched:
→ In-App: Supervisor sees alert card appear in real time.
→ Email: Supervisor receives email with snapshot.
→ Slack: #safety-welding channel notified.
Audit entry created: AlertCreated.

Scenario 2 — Zone Overcrowding

Welding Area max occupancy = 5.
Workers entering gradually.
Current count reaches 6.
OccupancyUpdated event: count=6, max=5.
Rules engine: overcrowding = True → rule fires (immediate, no duration).
Event published: OvercrowdingDetected.
Alert created: ALT-2026-0088, Severity=High.
Supervisor + Safety Officer notified.
Dashboard occupancy gauge turns Red.
Count drops to 4 → occupancy gauge returns to Green.
Alert remains Open until supervisor closes it.

Scenario 3 — Camera Offline

Camera CAM-005 loses RTSP connection.
Worker retries: 5s → 15s → 60s.
All retries fail.
Event published: CameraOfflineDetected (camera=CAM-005).
Alert created: ALT-2026-0089, Severity=High.
IT Admin notified via Email + In-App.
Dashboard: CAM-005 card turns Red with "Offline" badge.
Camera reconnects 10 minutes later.
Event published: CameraReconnected.
Alert auto-closed.
Dashboard: CAM-005 card returns to Green.

Scenario 4 — SLA Breach Escalation

Alert ALT-2026-0090 (Severity=High) created at 10:00 AM.
SLA: acknowledgement within 5 minutes.
10:05 AM: Alert still Open – no acknowledgement.
Escalation triggered.
Safety Officer receives escalation notification.
Audit entry: AlertEscalated.
10:07 AM: Safety Officer acknowledges → status = Acknowledged.
10:35 AM: SLA resolution time (30 min) approaching.
10:30 AM: Resolved by supervisor → status = Closed.
Alert closure time: 30 minutes (within SLA).

Scenario 5 — Zone Config Change (Hot-Swap)

Admin increases helmet threshold: 0.75 → 0.80 for Welding Area.
PUT /api/v1/config/zone/ZONE-WELDING
config.zone_configs updated: version 7 → 8.
ConfigUpdated event published: zone_id=ZONE-WELDING, version=8.
Worker 1 receives event.
Worker 1 local_version=7 < event.version=8 → apply new config.
Worker 1 updates in-memory threshold: helmet_threshold = 0.80.
Worker 1 continues stream – no restart.
Change takes effect within 5 seconds.
Config change logged to audit.

19. Success Criteria

Criterion	Target	Measurement
Helmet detection accuracy	≥ 90%	Test dataset evaluation
Occupancy count accuracy	≥ 95%	Benchmark test
Alert latency	≤ 5 seconds	End-to-end timing test
Camera offline detection	≤ 60 seconds	Disconnect test
Config propagation to workers	≤ 10 seconds	Timing test
API response time (p95)	≤ 200ms	Load test
System availability	≥ 99.5%	30-day uptime measurement
Camera support (initial)	50 cameras	Capacity test
Camera support (future)	200+ cameras	Architecture review
Zero data loss on worker crash	0 lost events	DLQ recovery test
Audit trail completeness	100%	Manual audit check
SLA breach escalation	100% automated	Test per severity

Criterion	Target	Measurement
Report generation	≤ 10 seconds	Timing test
WebSocket latency	≤ 1 second	Measured push delay

20. Future Roadmap

Phase 2 (3–6 months post-launch)

- Face recognition — link detected persons to worker identities
- Visitor management — track non-employees
- Attendance integration — auto-correlate camera detection with HRMS
- Gloves + shoes detection improvement
- SMS notification channel
- Mobile PWA (Progressive Web App) for supervisors
- Shift management — shift-wise violation analysis

Phase 3 (6–12 months)

- Fire and smoke detection (YOLO model extension)
- Fall detection
- SAP work order integration on violation
- Access control cross-reference (badge vs camera)
- Fire alarm system correlation
- Multi-language UI (Hindi, regional languages)
- Offline mode for dashboard (PWA cache)

Phase 4 (12–18 months)

- Forklift and vehicle detection
- Unsafe behavior detection (e.g., running in restricted areas)
- AI Risk Scoring — predict high-risk periods before incidents occur
- Digital safety audits — structured audit workflow in platform
- Predictive analytics — forecast violation trends
- Edge AI — on-camera inference for low-bandwidth factories
- Kubernetes deployment for 1000+ camera scale
- Multi-region deployment

End of Document — VisionGuard AI BRD v2.0 Enterprise Edition

21. Enterprise Hierarchy & Multi-Factory Architecture

21.1 Overview

VisionGuard AI is not a single-factory monitoring tool. It is an **Enterprise Platform** designed to centrally manage multiple factories under one organization (Enterprise), while keeping each factory's data, users, and configurations completely isolated.

Real-world example:

```
Enterprise: {Enterprise Name}
  ■■■ AC Factory ({City A})
  ■■■ Washing Machine Factory ({City B})
  ■■■ Refrigerator Factory ({City C})
  ■■■ E-commerce Factory ({City D})
```

All four factories are managed from a single VisionGuard AI deployment. The Head Office (HO) System Administrator can monitor all factories centrally. Each Factory Admin sees only their own factory.

21.2 5-Level Hierarchy

```
Level 1 → Enterprise {Enterprise Name}
Level 2 → Factory AC Factory
Level 3 → Department Assembly Department
Level 4 → Zone Zone A (Main Assembly Line)
Level 5 → Camera CAM-001, CAM-002
```

Why Department Level

The original BRD had only Factory → Zone. Department is a required addition because:

- Large factories have 10–20 zones — grouping under departments is essential
- Department Head needs isolated visibility (only their department)
- Alert routing must reach Department Head, not just Zone Supervisor
- Analytics must aggregate at department level
- Different departments have different PPE requirements (Welding ≠ Packaging)

21.3 Enterprise Structure Example

```
Enterprise: {Enterprise Name}
  ■
```



```

■■■■ Factory: AC Factory ({City A})
■ ■■■■ Department: Assembly
■ ■ ■■■■ Zone: Main Assembly Line (Max: 10) [3 cameras]
■ ■ ■■■■ Zone: Sub-Assembly Area (Max: 8) [2 cameras]
■ ■ ■■■■ Zone: Component Store (Max: 5) [1 camera]
■ ■■■■ Department: Welding
■ ■ ■■■■ Zone: Welding Bay A (Max: 5) [2 cameras]
■ ■ ■■■■ Zone: Welding Bay B (Max: 5) [2 cameras]
■ ■■■■ Department: Quality Control
■ ■ ■■■■ Zone: QC Lab (Max: 6) [2 cameras]
■ ■■■■ Department: Packaging
■ ■■■■ Zone: Packing Line (Max: 8) [2 cameras]
■
■■■■ Factory: Washing Machine Factory ({City B})
■ ■■■■ Department: Drum Assembly
■ ■■■■ Department: Final Assembly
■
■■■■ Factory: Refrigerator Factory ({City C})
■ ■■■■ ...
■
■■■■ Factory: E-commerce Factory ({City D})
■■■■ ...

```

21.4 User Roles — Updated for Enterprise

Role 1 — HO System Administrator (Head Office Admin)

Scope: Entire Enterprise — all factories.

Responsibilities:

- View enterprise-wide dashboard (all factories at a glance)
- Cross-factory analytics and comparison
- Manage all factories, departments, zones, cameras
- Manage all users across all factories
- View all alerts across all factories
- Generate enterprise-level compliance reports
- Access full audit trail across all factories
- Manage AI worker assignments
- Configure enterprise-wide settings

Key Differentiator: Only role that sees cross-factory data.

Role 2 — Factory Admin

Scope: Single assigned factory only.

Responsibilities:

- Manage departments, zones, cameras within their factory
- Manage users within their factory
- View factory-level dashboard, alerts, analytics
- Generate factory-level reports
- Configure factory-level detection rules

Restriction: Cannot see any data from other factories — even if they belong to the same enterprise.

Role 3 — Department Head

Scope: Single assigned department within a factory.

Responsibilities:

- Monitor all zones within their department
- View department-level dashboard and analytics
- Acknowledge and resolve alerts in their department
- View department compliance reports

Restriction: Cannot see zones or alerts outside their department.

Role 4 — Safety Officer

Scope: Single factory.

Responsibilities:

- Review and investigate all alerts in their factory
 - Generate safety compliance reports
 - View factory-wide analytics
-

Role 5 — Supervisor

Scope: Assigned zones only (within a department).

Responsibilities:

- Monitor assigned zones in real time
 - Acknowledge and resolve alerts for assigned zones
 - Take corrective actions
-

Role 6 — Viewer

Scope: Assigned dashboard scope (read-only).

21.5 Access Control Matrix

Feature	HO Admin	Factory Admin	Dept Head	Safety Officer	Supervisor	Viewer
Enterprise Dashboard	■	■	■	■	■	■
All Factories View	■	■	■	■	■	■
Own Factory Dashboard	■	■	■	■	■	■
Other Factory Data	■	■	■	■	■	■
Dept Dashboard	■	■	■ (own)	■	■	■
Zone Dashboard	■	■	■ (own dept)	■	■ (assigned)	■ (assigned)
User Management	■ (all)	■ (own factory)	■	■	■	■
Alert Management	■	■	■ (own dept)	■	■ (own zones)	■
Config Management	■	■	■	■	■	■
Enterprise Reports	■	■	■	■	■	■
Factory Reports	■	■	■	■	■	■
Audit Log	■ (all)	■ (own factory)	■	■ (own actions)	■	■

21.6 Enterprise Dashboard — HO Admin View

The Enterprise Dashboard is exclusive to HO Admin. It provides a bird's-eye view of all factories simultaneously.

Dashboard Panels

Factory Comparison Panel:

Factory	Safety Score	Active Alerts	Cameras Online	PPE Compliance	Violations Today
AC Factory	94%	2	18/18	96.2%	3
WM Factory	87%	7	22/24	89.1%	14
Fridge Factory	91%	1	12/12	93.5%	5
E-comm Factory	96%	0	8/8	98.1%	1

Enterprise Totals:

- Overall Safety Score: 92%
- Total Active Alerts: 10
- Total Cameras: 60 (58 online, 2 offline)
- Enterprise PPE Compliance: 94.2%

Charts (Enterprise Level):

- Safety score trend per factory (last 30 days — line chart, one line per factory)
- Violations by factory (bar chart — today)
- Worst performing zones across all factories (top 10 ranking)
- Alert volume heatmap (factory × day of week)

Drill-down: HO Admin can click any factory card to enter that factory's dashboard.

21.7 Factory Isolation Rules

FR-ENT-001: A Factory Admin shall only see data belonging to their assigned factory.

FR-ENT-002: System shall enforce factory isolation at the database query level — not just at the UI level.

FR-ENT-003: API shall return HTTP 403 if a Factory Admin attempts to access another factory's data — even within the same enterprise.

FR-ENT-004: All alert notifications shall only be sent to users within the correct factory scope.

FR-ENT-005: Reports generated by a Factory Admin shall only contain data from their factory.

FR-ENT-006: Audit log entries shall be visible to Factory Admin only for their own factory's actions.

21.8 Cross-Factory Analytics (HO Admin Only)

HO Admin has access to analytics that span all factories:

Cross-Factory PPE Comparison:

- Helmet compliance % per factory — bar chart
- Vest compliance % per factory — bar chart
- Which factory has the worst PPE compliance this month?

Cross-Factory Alert Analysis:

- Alert volume per factory per week
- Average alert resolution time per factory
- Which factory has the most SLA breaches?

Safety Score Ranking:

- Factory ranking by safety score (best to worst)
- Month-over-month improvement per factory
- Which department across all factories has the most violations?

Enterprise KPI Summary (Monthly Report):

- Total violations across all factories
- Total incidents
- Enterprise-wide PPE compliance %
- Enterprise safety score
- Factory-wise breakdown

21.9 Multi-Enterprise Support

VisionGuard AI is designed to be supplied to multiple enterprise clients. Each enterprise client gets complete data isolation:

```
Enterprise 1: {Enterprise Name}
→ enterprise_id: uuid-enterprise-1
→ 4 factories, 60 cameras

Enterprise 2: {Enterprise Name 2}
→ enterprise_id: uuid-enterprise-2
→ 6 factories, 350 cameras

Enterprise 3: {Enterprise Name 3}
→ enterprise_id: uuid-enterprise-3
→ 3 factories, 150 cameras
```

Isolation guarantee:

- {Enterprise Name}'s HO Admin cannot access Tata's data
- Enforced via PostgreSQL Row Level Security (RLS)
- Every table carries `enterprise_id` — every query is scoped to it

Super Admin (VisionGuard Internal Team)

A `SUPER_ADMIN` role exists for the VisionGuard platform operations team:

Capabilities:

- Create and manage enterprise accounts
- Activate / suspend enterprise access
- View platform health across all enterprises
- Access any enterprise for support purposes (audit-logged)
- Manage platform-wide configurations

Security:

- Accessed via separate `/superadmin/` URL
 - Requires MFA
 - Every Super Admin action is logged in a separate immutable platform audit log
 - Super Admin actions are never visible in client-facing audit logs
-

21.10 New Functional Requirements

FR-ENT-001 to FR-ENT-010 — Enterprise Management

FR-ENT-001: System shall support multiple enterprises on a single platform deployment.

FR-ENT-002: Each enterprise shall have complete data isolation from all other enterprises.

FR-ENT-003: HO Admin shall see all factories within their enterprise from a single dashboard.

FR-ENT-004: Factory Admin shall only see their own factory's data — no cross-factory visibility.

FR-ENT-005: System shall enforce access scope at the API middleware level on every request.

FR-ENT-006: Alert notifications shall be scoped to the correct factory — a Factory Admin from AC Factory shall not receive alerts from WM Factory.

FR-ENT-007: Reports shall be scoped to the requesting user's access level automatically.

FR-ENT-008: Audit log shall be scoped per factory for Factory Admin, and enterprise-wide for HO Admin.

FR-ENT-009: Super Admin shall be able to create, activate, and suspend enterprise accounts.

FR-ENT-010: Super Admin access to any enterprise shall be logged in a separate immutable platform audit log.

FR-DEPT-001 to FR-DEPT-005 — Department Management

FR-DEPT-001: Admin shall create departments within a factory with:

- Department Name
- Department Code (unique within factory)
- Department Head (linked user)
- Description
- Status (Active / Inactive)

FR-DEPT-002: Department shall group multiple zones.

FR-DEPT-003: Department Head shall receive alerts for all zones within their department.

FR-DEPT-004: Department dashboard shall show:

- All zones in department with live occupancy
- Active alerts for department
- Department PPE compliance %
- Department safety score

FR-DEPT-005: Analytics shall be available at department level:

- Violations by zone within department
- Department compliance trend
- Department vs factory average comparison

21.11 Updated Database Tables

enterprises (new)

```
id UUID PK
name VARCHAR ({Enterprise Name})
code VARCHAR UNIQUE (HAIER-IN)
industry VARCHAR
contact_person VARCHAR
contact_email VARCHAR
status ENUM (Active, Inactive, Suspended)
created_on TIMESTAMPTZ
created_by UUID
```

departments (new)

```
id UUID PK
enterprise_id FK → enterprises
factory_id FK → factories
name VARCHAR
code VARCHAR (unique within factory)
head_user_id FK → users
description TEXT
status ENUM
created_on TIMESTAMPTZ
created_by UUID
modified_by UUID
version INT DEFAULT 1
deleted_at TIMESTAMPTZ
```

factories (updated)

```
enterprise_id FK → enterprises ← ADDED
```

zones (updated)

```
department_id FK → departments ← ADDED
```

All other tables (updated)

enterprise_id FK → enterprises ← ADDED to every table
for RLS enforcement

21.12 Updated Workflow Scenarios

Scenario 6 — HO Admin Cross-Factory Monitoring

HO Admin logs in to VisionGuard AI.
Enterprise dashboard loads showing all 4 factories.
WM Factory shows: Safety Score 87%, Active Alerts 7, 2 cameras offline.
HO Admin notices WM Factory is underperforming vs others.
HO Admin clicks WM Factory → enters WM Factory dashboard.
Sees: 7 open alerts, Welding Dept has 5 of them.
HO Admin contacts WM Factory Admin via platform notification.
WM Factory Admin acknowledges and begins investigation.
HO Admin returns to enterprise dashboard – monitors resolution.

Scenario 7 — Factory Admin Isolation

AC Factory Admin logs in.
Dashboard shows only AC Factory data.
AC Factory Admin tries to access:
GET /api/v1/factories/uuid-factory-2/alerts
System returns: 403 Forbidden
Audit log entry created: UnauthorizedAccess
AC Factory Admin has no visibility into WM Factory.

Scenario 8 — Department Head Alert Routing

Worker in Welding Bay A (AC Factory > Welding Dept > Zone A) missing helmet.
Alert created with full hierarchy:
enterprise_id = uuid-enterprise-1
factory_id = uuid-factory-1
department_id = uuid-welding-dept
zone_id = uuid-zone-a

Notifications sent:
Zone Supervisor (Welding Bay A) → In-App + Email
Department Head (Welding Dept) → In-App + Email
Factory Safety Officer → In-App
HO Admin → Visible on enterprise dashboard

AC Factory Admin → sees it on factory dashboard
WM Factory Admin → does NOT receive this alert (different factory)

End of Enterprise Hierarchy & Multi-Factory Architecture Section

22. Dynamic Branding Requirements

22.1 Core Rule

No company name, logo, or brand asset shall be hardcoded anywhere in the VisionGuard AI platform.

This applies to:

- All UI screens (dashboard, login, alerts, reports, settings)
- All email notifications
- All PDF and Excel reports
- All export filenames
- All audit log display text
- All notification templates

Every piece of branding must be loaded dynamically from the database at runtime, based on the logged-in user's enterprise.

22.2 Enterprise Branding Fields

Field	Type	Purpose
`name`	VARCHAR	Shown in UI headers, emails, PDF reports
`code`	VARCHAR	Used in export filenames
`logo_url`	VARCHAR	MinIO path — served as pre-signed URL
`favicon_url`	VARCHAR	Browser tab icon
`primary_color`	VARCHAR	Hex color for UI theme
`secondary_color`	VARCHAR	Secondary brand color
`tagline`	VARCHAR	Optional subtitle on login page

22.3 Functional Requirements

FR-BRD-001: System shall display enterprise name dynamically from database on all UI screens.

FR-BRD-002: System shall display enterprise logo dynamically from MinIO storage on login page, dashboard header, and all PDF reports.

- FR-BRD-003:** System shall apply enterprise primary color as UI theme color dynamically.
- FR-BRD-004:** All email notification subjects and bodies shall use enterprise name from database — no hardcoded company names.
- FR-BRD-005:** All generated PDF reports shall include enterprise name and logo from database in header and footer.
- FR-BRD-006:** All export filenames shall use enterprise code from database (e.g. `{enterprise_code}_report_2026.pdf`).
- FR-BRD-007:** Enterprise Admin shall be able to upload and update logo via platform settings.
- FR-BRD-008:** Super Admin shall be able to set branding for any enterprise on their behalf.
- FR-BRD-009:** Logo shall be stored in MinIO under `logos/{enterprise_id}/logo.png` and served via pre-signed URL.
- FR-BRD-010:** Browser tab favicon shall be set dynamically from `enterprise.favicon_url`.

22.4 Where Branding Appears

Location	Dynamic Value
Browser tab	<code>{enterprise.name} — VisionGuard AI</code>
Login page logo	<code>enterprise.logo_url</code>
Login page heading	<code>enterprise.name</code>
Dashboard header	<code>enterprise.logo_url</code> + <code>enterprise.name</code>
UI accent color	<code>enterprise.primary_color</code>
Email subject	<code>[{enterprise.name}] Alert — {violation_type}</code>
Email header	<code>enterprise.logo_url</code> + <code>enterprise.name</code>
PDF report header	<code>enterprise.name</code> + <code>enterprise.logo_url</code>
PDF report footer	<code>enterprise.name</code>
Export filename	<code>{enterprise.code}_safety_report_2026.pdf</code>

End of Dynamic Branding Requirements Section

23. First Time Setup & Onboarding Flow

23.1 Overview

When VisionGuard AI is deployed for a new enterprise client, the system must guide administrators through a structured onboarding process. No factory, department, zone, or camera data exists initially — the platform must be configured before monitoring can begin.

There are two onboarding actors:

Actor	Role	Responsibility
Super Admin	VisionGuard internal team	Creates enterprise + first HO Admin user
HO Admin (Enterprise Admin)	Client's head office admin	Creates factories, departments, zones, cameras

23.2 Complete Onboarding Flow

PHASE 1 – Super Admin Setup (VisionGuard Team)

Step 1 → Super Admin logs in
(Platform default credentials, MFA required)


```
Step 2 → Creates Enterprise account
Fields: Enterprise Name, Code, Industry,
Logo, Primary Color, Contact Person,
Contact Email, Status
```


Step 3 → Creates first H0 Admin user for enterprise

Fields: Full Name, Email, Temporary Password

Role: HO_ADMIN

Step 4 → System sends Welcome Email to HO Admin

Contains: Login URL, Temporary Password,
Password reset link (expires 24h)

PHASE 2 – H0 Admin First Login

Step 5 → H0 Admin receives Welcome Email

Step 6 → H0 Admin clicks login link
Prompted to set new password (mandatory)

Step 7 → System detects: `is_first_login = true`

```
setup_completed = false
```

```
Setup_Completed = false
Redirects to Setup Wizard automatically
```


Journal Pre-proof

PHASE 3 – Setup Wizard (H0 Admin completes)

```
Step 8 → Screen 1: Welcome Screen
Step 9 → Screen 2: Create First Factory
Step 10 → Screen 3: Create First Department
Step 11 → Screen 4: Create First Zone + PPE Config
Step 12 → Screen 5: Add First Camera (RTSP URL)
Step 13 → Screen 6: Summary + Launch Dashboard
↓
Step 14 → setup_completed = true saved to DB
↓
Step 15 → AI Worker auto-assigned to cameras
↓
Step 16 → Dashboard goes live ■
```

23.3 Setup Wizard — Screen Details

Screen 1 — Welcome

[illegible]

Screen 2 — Create First Factory

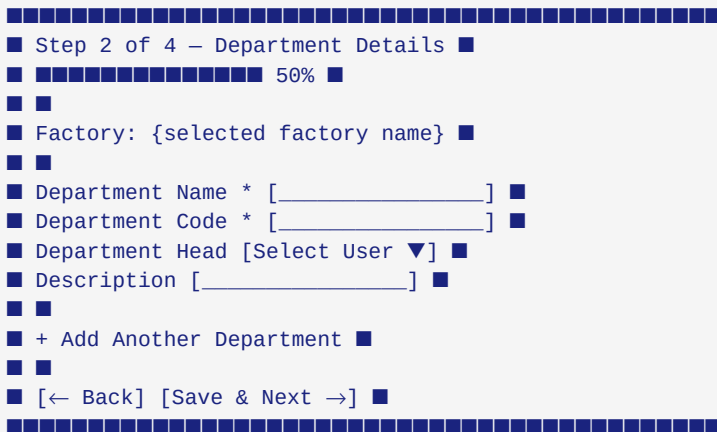
[illegible]



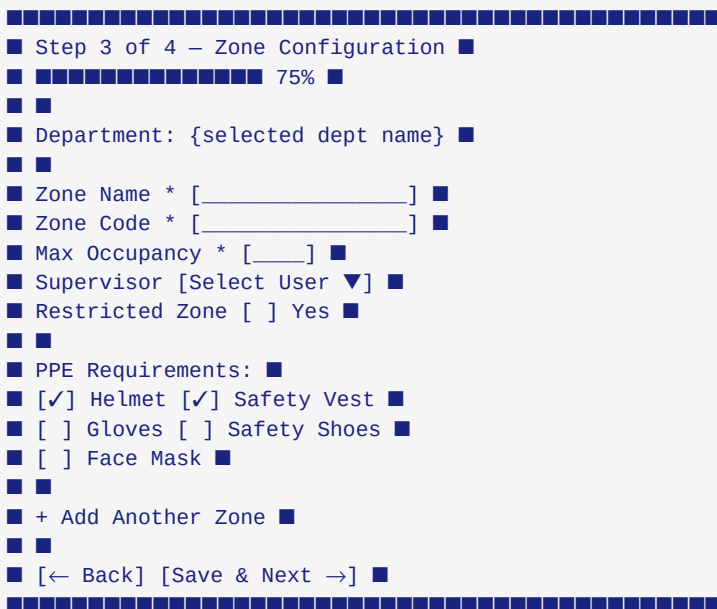
Validation:

- Factory Name: required, max 100 chars
- Factory Code: required, unique within enterprise, alphanumeric + hyphen only
- Plant Head: optional at this stage (can be assigned later)

Screen 3 — Create First Department



Screen 4 — Create First Zone + PPE Config



Screen 5 — Add First Camera

```

■ Step 4 of 4 – Camera Setup ■
■ ████████████████████ 100% ■
■ ■
■ Zone: {selected zone name} ■
■ ■
■ Camera Name * [ _____ ] ■
■ Camera Code * [ _____ ] ■
■ RTSP URL * [ _____ ] ■
■ Camera Type [Fixed ▼] ■
■ Position [ _____ ] ■
■ ■
■ [Test Connection] ← validates RTSP URL ■
■ ■
■ + Add Another Camera ■
■ ■
■ [← Back] [Complete Setup →] ■

```

RTSP Test Connection:

- System attempts to connect to RTSP URL
- Shows:  Connected (FPS: 25) or  Connection Failed
- Admin can still save even if test fails (camera may be offline temporarily)

Screen 6 — Setup Complete

[illegible]

23.4 First Login Detection

System tracks two flags per user:

Field	Type	Default	Meaning
`is_first_login`	BOOLEAN	true	Has user logged in before?
`setup_completed`	BOOLEAN	false	Has setup wizard been completed?

Behavior:

```
User logs in
↓
is_first_login = true?
→ Force password change screen
→ Set is_first_login = false
↓
setup_completed = false AND role = HO_ADMIN?
→ Redirect to Setup Wizard
↓
setup_completed = true?
→ Normal dashboard
```

23.5 Post-Setup — What HO Admin Can Do Next

After setup wizard completes, HO Admin can:

- Add more factories (no limit based on subscription)
- Add more departments to any factory
- Add more zones to any department
- Add more cameras to any zone
- Create additional users and assign roles
- Upload enterprise logo and configure branding
- Configure detection rules per zone
- Set up notification preferences
- Invite Factory Admins for each factory

23.6 Factory Admin Onboarding (After HO Admin Creates Them)

```
HO Admin creates Factory Admin user
↓
System sends Welcome Email to Factory Admin
↓
Factory Admin logs in → forced password change
↓
is_first_login = false (no wizard for Factory Admin)
↓
Factory Admin lands on their Factory Dashboard
↓
If factory has no departments/zones yet →
```

Show banner: "Your factory has no zones configured.
Contact your HO Admin or add zones now."

23.7 Functional Requirements — Onboarding

FR-ONB-001: Super Admin shall be able to create a new enterprise with name, code, logo, colors, and contact details.

FR-ONB-002: Super Admin shall create the first HO Admin user for each enterprise during enterprise creation.

FR-ONB-003: System shall send a Welcome Email to newly created HO Admin with a one-time password setup link (expires in 24 hours).

FR-ONB-004: System shall detect first login (`is_first_login = true`) and force password change before any other action.

FR-ONB-005: System shall redirect HO Admin to Setup Wizard automatically if `setup_completed = false`.

FR-ONB-006: Setup Wizard shall guide HO Admin through: Factory → Department → Zone → Camera in sequence.

FR-ONB-007: Setup Wizard shall show progress indicator (step X of 4).

FR-ONB-008: Each step of the Setup Wizard shall validate inputs before proceeding to the next step.

FR-ONB-009: Camera screen shall provide a "Test Connection" button to validate RTSP URL before saving.

FR-ONB-010: Setup Wizard shall allow adding multiple factories, departments, zones, and cameras before completing.

FR-ONB-011: On Setup Wizard completion, system shall set `setup_completed = true` and redirect to dashboard.

FR-ONB-012: AI Worker shall be auto-assigned to cameras upon setup completion.

FR-ONB-013: Setup Wizard progress shall be saved at each step — if admin closes browser midway, they resume from last completed step on next login.

FR-ONB-014: HO Admin shall be able to re-access Setup Wizard from Admin Settings if they wish to add more entities post-setup.

FR-ONB-015: Factory Admin shall not see Setup Wizard — they land directly on their factory dashboard.

23.8 Database Fields — Onboarding Tracking

Updated: identity.users

```
is_first_login BOOLEAN DEFAULT true
setup_completed BOOLEAN DEFAULT false
password_changed_at TIMESTAMPTZ
invited_by FK → identity.users
invited_at TIMESTAMPTZ
```

New: onboarding.setup_progress

```
id UUID PK
user_id FK → identity.users
enterprise_id FK → enterprises
last_completed_step INT DEFAULT 0
factory_id FK → factories (nullable – step 1 result)
department_id FK → departments (nullable – step 2 result)
zone_id FK → zones (nullable – step 3 result)
camera_id FK → cameras (nullable – step 4 result)
completed_at TIMESTAMPTZ
```

This allows resuming wizard from last completed step if admin closes browser midway.

23.9 Welcome Email Template

```
Subject: Welcome to {enterprise.name} – VisionGuard AI

[Enterprise Logo]
{enterprise.name} Safety Monitoring Platform

Hello {user.name},

Your VisionGuard AI account has been created.

Login URL: {platform_url}/login
Email: {user.email}
Password: {temporary_password}

This password expires in 24 hours.
You will be prompted to set a new password on first login.

If you did not expect this email, please contact your system administrator.

– VisionGuard AI Platform
```

End of First Time Setup & Onboarding Flow Section

24. Security Enhancements

24.1 Two Factor Authentication (2FA)

FR-SEC-001: System shall support optional 2FA for all user roles via TOTP (Google Authenticator / Authy).

FR-SEC-002: 2FA shall be mandatory for Super Admin and HO Admin roles.

FR-SEC-003: User shall be able to enable/disable 2FA from their profile settings (except mandatory roles).

FR-SEC-004: On 2FA enable, system shall show QR code for authenticator app setup.

FR-SEC-005: Backup codes (8 codes) shall be generated on 2FA setup for account recovery.

24.2 Session Management

FR-SEC-006: System shall auto-logout idle users after configurable timeout (default: 30 minutes).

FR-SEC-007: Admin shall configure session timeout per role:

Role	Default Timeout
Super Admin	15 minutes
HO Admin	30 minutes
Factory Admin	60 minutes
Supervisor	120 minutes
Viewer	240 minutes

FR-SEC-008: User shall see a warning popup 2 minutes before session expiry with option to extend.

FR-SEC-009: On session expiry, user shall be redirected to login page with message "Session expired. Please login again."

24.3 IP Whitelisting

FR-SEC-010: Enterprise Admin shall configure allowed IP ranges for their enterprise.

FR-SEC-011: Login attempts from non-whitelisted IPs shall be blocked and logged to audit.

FR-SEC-012: IP whitelist shall support individual IPs and CIDR ranges (e.g. 192.168.1.0/24).

FR-SEC-013: Super Admin shall bypass IP whitelist (for support access).

24.4 Password Expiry

FR-SEC-014: System shall enforce password expiry after configurable period (default: 90 days).

FR-SEC-015: User shall receive email reminder 7 days before password expiry.

FR-SEC-016: On password expiry, user shall be forced to change password before accessing platform.

FR-SEC-017: Last 5 passwords shall not be reusable.

25. Alert Management Enhancements

25.1 Full Escalation Matrix

FR-ALT-020: System shall support a configurable multi-level escalation matrix per severity:

Level	Trigger	Notify
Level 1	Alert created	Zone Supervisor
Level 2	No ack in 5 min (High)	Department Head
Level 3	No ack in 15 min (High)	Safety Officer
Level 4	No ack in 30 min (High)	Factory Admin
Level 5	No ack in 60 min (High)	HO Admin

FR-ALT-021: Escalation matrix shall be fully configurable per enterprise per severity level.

FR-ALT-022: Each escalation event shall be logged to audit trail.

FR-ALT-023: Escalation shall stop as soon as alert is acknowledged at any level.

25.2 Alert Bulk Actions

FR-ALT-024: Supervisor shall be able to select multiple alerts and perform bulk actions:

- Bulk Acknowledge
- Bulk Assign (to a user)
- Bulk Close (with mandatory comment)

FR-ALT-025: Bulk actions shall be logged individually in alert history (each alert gets its own history entry).

FR-ALT-026: Bulk action limit: maximum 100 alerts at once.

25.3 False Positive Marking

FR-ALT-027: Supervisor shall be able to mark an alert as "False Positive" with a reason.

FR-ALT-028: False positive alerts shall be excluded from compliance % calculations.

FR-ALT-029: False positive rate shall be tracked per camera — high false positive rate triggers AI model review recommendation.

FR-ALT-030: False positive reasons shall be stored and used for model improvement reporting.

False positive reasons (configurable):

- Poor lighting
- Camera angle obstruction
- Reflection / glare
- Worker bending (helmet not visible)
- Other (free text)

25.4 Alert Snooze

FR-ALT-031: Supervisor shall be able to snooze alerts for a specific camera for a defined period:

- 15 minutes
- 30 minutes
- 1 hour
- 2 hours
- Custom duration

FR-ALT-032: Snooze reason shall be mandatory.

FR-ALT-033: During snooze period, no new alerts of the same type shall be created for that camera.

FR-ALT-034: Snooze shall auto-expire at the configured time and alerts shall resume.

FR-ALT-035: Snooze actions shall be logged to audit trail.

25.5 Alert Templates (Resolution Notes)

FR-ALT-036: Admin shall configure common resolution note templates:

- "Worker warned and PPE provided"
- "Worker removed from zone"
- "Camera obstruction cleared"
- "False alarm — lighting issue"

FR-ALT-037: Supervisor shall select a template when closing an alert (or write custom note).

26. Dashboard Enhancements

26.1 Full Screen Mode

FR-DASH-010: Dashboard shall support full screen mode for control room TV displays.

FR-DASH-011: Full screen mode shall auto-rotate between:

- Enterprise overview
- Factory 1 dashboard
- Factory 2 dashboard
- Alert feed

FR-DASH-012: Rotation interval shall be configurable (default: 30 seconds per screen).

26.2 Dark Mode

FR-DASH-013: Platform shall support Dark Mode toggle.

FR-DASH-014: User preference (dark/light) shall be saved per user in DB.

FR-DASH-015: Control room full screen mode shall default to Dark Mode.

26.3 Activity Feed

FR-DASH-016: Dashboard shall show a real-time Activity Feed panel showing last 20 events:

```
[2 min ago] Alert resolved – Helmet Missing – Zone A – by Ravi Kumar  
[5 min ago] Zone B occupancy exceeded (8/5)  
[8 min ago] Camera CAM-003 came back online  
[12 min ago] Config updated – Welding Zone threshold changed  
[15 min ago] New alert – Vest Missing – Zone C
```

FR-DASH-017: Activity feed shall update in real time via WebSocket.

FR-DASH-018: Activity feed items shall be clickable — click navigates to relevant entity.

26.4 Dashboard Auto-Refresh

FR-DASH-019: User shall configure dashboard auto-refresh interval:

- Real-time (WebSocket — default)
- Every 30 seconds
- Every 1 minute

- Every 5 minutes

27. Notification Enhancements

27.1 SMS Notifications

FR-NOT-010: System shall support SMS notifications via configurable SMS gateway (Twilio / local provider).

FR-NOT-011: SMS shall be sent for Critical and High severity alerts only (to avoid spam).

FR-NOT-012: SMS template shall be concise (160 chars max):

```
{enterprise_code} {severity} Alert: {violation_type} at {zone_name}.  
View: {short_url}
```

27.2 Notification Digest

FR-NOT-013: User shall configure notification digest instead of individual notifications:

- Real-time (default)
- Every 15 minutes summary
- Every 1 hour summary
- Daily digest (end of day)

FR-NOT-014: Digest shall group alerts by severity and show count:

```
Last 1 hour summary:  
Critical: 0  
High: 3 (2 Helmet Missing, 1 Overcrowding)  
Medium: 7  
Low: 2  
View all: {dashboard_url}
```

FR-NOT-015: Critical alerts shall always be sent immediately regardless of digest setting.

27.3 Do Not Disturb (DND)

FR-NOT-016: User shall configure DND hours (e.g. 10 PM — 6 AM).

FR-NOT-017: During DND, only Critical alerts shall bypass DND and notify immediately.

FR-NOT-018: Non-critical alerts during DND shall be queued and delivered as a digest at DND end time.

27.4 Notification Read/Unread Tracking

FR-NOT-019: In-app notifications shall show unread count badge on notification bell icon.

FR-NOT-020: User shall mark individual notifications as read or mark all as read.

FR-NOT-021: Unread notifications older than 7 days shall be auto-marked as read.

28. Reports Enhancements

28.1 Scheduled Email Reports

FR-RPT-010: Admin shall configure scheduled reports with:

- Report type
- Frequency (Daily / Weekly / Monthly)
- Delivery time
- Recipients (email list)
- Format (PDF / Excel / Both)

FR-RPT-011: System shall auto-generate and email scheduled reports at configured time.

FR-RPT-012: Failed report delivery shall retry 3 times and notify admin on all retries fail.

28.2 Custom Date Range Reports

FR-RPT-013: User shall generate reports for any custom date range (not just daily/weekly/monthly).

FR-RPT-014: Maximum date range for a single report: 90 days.

28.3 Comparative Reports

FR-RPT-015: System shall generate comparative reports:

- This month vs Last month
- This week vs Last week
- Factory A vs Factory B (HO Admin only)
- Zone A vs Zone B (within same factory)

FR-RPT-016: Comparative report shall show % change with up/down indicators.

28.4 Zone-wise & Camera-wise Reports

FR-RPT-017: User shall generate reports filtered by specific zone.

FR-RPT-018: User shall generate reports filtered by specific camera.

28.5 Data Export (CSV)

FR-RPT-019: User shall export raw data as CSV for:

- Violations list
- Occupancy logs
- Alert history
- Audit log

FR-RPT-020: CSV export shall respect user's access scope (Factory Admin gets only their factory's data).

FR-RPT-021: CSV filenames shall use enterprise code: `{enterprise_code}_violations_2026-01.csv`

29. Configuration Enhancements

29.1 Config History

FR-CFG-010: System shall maintain full history of zone config changes:

- What changed (old value → new value)
- Who changed it
- When it was changed

FR-CFG-011: Admin shall view config history per zone from settings page.

FR-CFG-012: Admin shall restore a previous config version with one click.

29.2 Config Copy

FR-CFG-013: Admin shall copy zone configuration from one zone to another.

FR-CFG-014: Admin shall copy zone configuration to multiple zones at once (bulk copy).

FR-CFG-015: Before copy, system shall show preview of what will change.

29.3 Config Templates

FR-CFG-016: Admin shall save a zone config as a named template:

- "Welding Zone Standard"
- "Chemical Storage High Security"
- "General Assembly Standard"

FR-CFG-017: Admin shall apply a saved template to any zone.

FR-CFG-018: Templates shall be available enterprise-wide (not factory-specific).

29.4 Bulk Config Update

FR-CFG-019: Admin shall select multiple zones and update a specific setting for all at once.

Example: Update helmet threshold from 0.75 to 0.80 for all welding zones across all factories.

FR-CFG-020: Bulk config update shall show confirmation with list of affected zones before applying.

30. User Management Enhancements

30.1 User Activity Log

FR-USR-010: System shall track per-user activity:

- Last login timestamp
- Last active timestamp
- Last action performed
- Total alerts resolved (count)
- Total logins (count)

FR-USR-011: Admin shall view user activity report from user management page.

30.2 Bulk User Import

FR-USR-012: Admin shall import users via CSV file with columns:

```
name, email, role, factory_code, department_code, zone_codes
```

FR-USR-013: System shall validate CSV before import and show error report for invalid rows.

FR-USR-014: On successful import, system shall send Welcome Email to all imported users.

FR-USR-015: Maximum bulk import: 500 users per file.

30.3 User Groups

FR-USR-016: Admin shall create user groups (e.g. "Welding Supervisors", "Night Shift Team").

FR-USR-017: Notifications shall be sendable to a user group instead of individual users.

FR-USR-018: Alert routing shall support assigning to a user group (any member can acknowledge).

30.4 Temporary Access

FR-USR-019: Admin shall grant temporary access to a user with auto-expiry date.

FR-USR-020: On expiry, account shall be auto-deactivated.

FR-USR-021: Admin shall receive email reminder 1 day before temporary access expires.

31. Camera & Infrastructure Enhancements

31.1 On-Demand Snapshot

FR-CAM-010: User shall request an on-demand snapshot from any active camera.

FR-CAM-011: On-demand snapshot shall be captured by AI worker and stored in MinIO within 5 seconds.

FR-CAM-012: Snapshot shall be viewable directly in dashboard and downloadable.

31.2 Camera Maintenance Mode

FR-CAM-013: Admin shall put a camera into "Maintenance Mode":

- No alerts generated during maintenance
- No offline alerts triggered
- Dashboard shows "Under Maintenance" badge

FR-CAM-014: Maintenance mode shall require:

- Start time
- Expected end time
- Reason

FR-CAM-015: On maintenance end time, camera shall auto-exit maintenance mode.

FR-CAM-016: Maintenance mode actions shall be logged to audit trail.

31.3 Camera Grouping / Views

FR-CAM-017: User shall create custom camera groups (e.g. "North Wing Cameras", "Entry Gates").

FR-CAM-018: Dashboard shall support multi-camera grid view (2×2, 3×3, 4×4) for grouped cameras.

FR-CAM-019: Full screen multi-camera view shall be available for control room use.

31.4 Bulk Camera Import

FR-CAM-020: Admin shall import cameras via CSV:

```
name, code, rtsp_url, zone_code, camera_type, position
```

FR-CAM-021: System shall validate RTSP URL format for all rows before import.

FR-CAM-022: Maximum bulk import: 200 cameras per file.

32. Shift Management

32.1 Shift Configuration

FR-SHF-001: Admin shall configure shifts per factory:

Field	Example
Shift Name	Morning Shift
Start Time	06:00 AM
End Time	02:00 PM
Days	Monday to Saturday

FR-SHF-002: Factory shall support up to 3 shifts (Morning / Evening / Night).

FR-SHF-003: Shifts shall be configurable per factory — different factories can have different shift timings.

32.2 Shift-wise Analytics

FR-SHF-004: All violations, alerts, and occupancy data shall be tagged with the active shift at time of detection.

FR-SHF-005: Analytics shall show shift-wise breakdown:

- Violations per shift

- Compliance % per shift
- Alert volume per shift
- Peak occupancy per shift

FR-SHF-006: Dashboard shall show currently active shift name and time remaining.

FR-SHF-007: System shall identify highest-risk shift (most violations) and highlight it in analytics.

32.3 Shift-wise Reports

FR-SHF-008: Reports shall be generatable per shift.

FR-SHF-009: Daily report shall include shift-wise breakdown section.

33. Camera Maintenance Management

33.1 Maintenance Schedule

FR-MNT-001: Admin shall schedule preventive maintenance for cameras:

- Camera
- Scheduled date
- Maintenance type (Cleaning / Calibration / Hardware check)
- Assigned technician
- Notes

FR-MNT-002: System shall send reminder notifications 24 hours before scheduled maintenance.

FR-MNT-003: Technician shall mark maintenance as completed with:

- Completion timestamp
- Work performed (notes)
- Next maintenance due date

33.2 Maintenance History

FR-MNT-004: System shall maintain full maintenance history per camera.

FR-MNT-005: Camera detail page shall show:

- Last maintenance date
- Next maintenance due
- Maintenance history log

FR-MNT-006: Cameras overdue for maintenance (> 90 days since last maintenance) shall show warning badge.

34. Announcements / Notice Board

34.1 Factory Announcements

FR-ANN-001: Factory Admin / HO Admin shall post announcements visible to all users in scope:

Scope	Who Sees It
Enterprise-wide	All users in enterprise
Factory-wide	All users in that factory
Department-wide	All users in that department

FR-ANN-002: Announcement fields:

- Title
- Message (rich text)
- Scope (Enterprise / Factory / Department)
- Priority (Normal / Urgent)
- Expiry date (auto-hides after expiry)

FR-ANN-003: Urgent announcements shall appear as a banner on top of dashboard.

FR-ANN-004: Normal announcements shall appear in a Notice Board panel on dashboard.

FR-ANN-005: Users shall acknowledge announcements (read receipt).

35. API Access for Clients

35.1 Client API Keys

FR-API-001: Enterprise Admin shall generate API keys for external integrations.

FR-API-002: API key shall have configurable permissions (read-only / read-write per module).

FR-API-003: API key shall have configurable expiry date.

FR-API-004: All API key usage shall be logged (endpoint, timestamp, IP).

FR-API-005: Admin shall revoke API keys instantly.

35.2 Webhook Support

FR-API-006: Enterprise Admin shall configure webhooks for events:

- Alert Created
- Alert Resolved
- Violation Detected
- Camera Offline

FR-API-007: Webhook payload shall be JSON with full event details.

FR-API-008: Failed webhook deliveries shall retry 3 times.

FR-API-009: Webhook delivery log shall be viewable from settings.

36. Shift Management DB Tables

shifts

```
id UUID PK
factory_id FK → factories
enterprise_id FK → enterprises
name VARCHAR (Morning Shift)
start_time TIME (06:00:00)
end_time TIME (14:00:00)
days JSONB (["MON", "TUE", "WED", "THU", "FRI", "SAT"])
status ENUM (Active / Inactive)
created_on TIMESTAMPTZ
created_by FK → users
```

camera_maintenance

```
id UUID PK
camera_id FK → cameras
enterprise_id FK → enterprises
scheduled_date DATE
maintenance_type ENUM (Cleaning / Calibration / Hardware)
assigned_to FK → users
status ENUM (Scheduled / InProgress / Completed / Overdue)
notes TEXT
completed_at TIMESTAMPTZ
completed_by FK → users
completion_notes TEXT
next_due_date DATE
created_on TIMESTAMPTZ
```

```
created_by FK → users
```

announcements

```
id UUID PK
enterprise_id FK → enterprises
factory_id FK → factories (nullable – enterprise-wide if null)
department_id FK → departments (nullable)
title VARCHAR
message TEXT
priority ENUM (Normal / Urgent)
scope ENUM (Enterprise / Factory / Department)
expires_at TIMESTAMPTZ
created_by FK → users
created_on TIMESTAMPTZ
```

announcement_reads

```
id UUID PK
announcement_id FK → announcements
user_id FK → users
read_at TIMESTAMPTZ
```

alert_snooze

```
id UUID PK
camera_id FK → cameras
zone_id FK → zones
violation_type ENUM
snoozed_by FK → users
snooze_reason TEXT
snoozed_until TIMESTAMPTZ
created_on TIMESTAMPTZ
```

config_history

```
id UUID PK
zone_id FK → zones
enterprise_id FK → enterprises
changed_by FK → users
old_config JSONB
new_config JSONB
change_reason TEXT
changed_at TIMESTAMPTZ
```

config_templates

```
id UUID PK
enterprise_id FK → enterprises
name VARCHAR (Welding Zone Standard)
description TEXT
config_data JSONB
created_by FK → users
created_on TIMESTAMPTZ
```

user_groups

```
id UUID PK
enterprise_id FK → enterprises
factory_id FK → factories
name VARCHAR (Night Shift Supervisors)
description TEXT
created_by FK → users
created_on TIMESTAMPTZ
```

user_group_members

```
id UUID PK
group_id FK → user_groups
user_id FK → users
added_by FK → users
added_on TIMESTAMPTZ
```

api_keys

```
id UUID PK
enterprise_id FK → enterprises
name VARCHAR
key_hash VARCHAR (SHA-256, never plaintext)
permissions JSONB
expires_at TIMESTAMPTZ
last_used_at TIMESTAMPTZ
status ENUM (Active / Revoked)
created_by FK → users
created_on TIMESTAMPTZ
```

End of Enhancement Sections